Advanced Image Processing Methods
Homework 4 Report

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Goal

Estimate a relative depth (disparity) map from an uncalibrated stereo image pair using fundamental matrix estimation, uncalibrated rectification, and Semi-Global Block Matching.

Method

1. **Image Loading**: Both views (1920×1080) are loaded as grayscale with `cv2.imread`.
2. **Keypoint Detection & Matching**: SIFT is used for detection — its 128-D float descriptors yield more accurate sub-pixel matches than ORB, which is important since fundamental matrix quality depends directly on match accuracy. Descriptors are matched via FLANN (KD-Tree), filtered with Lowe's ratio test (0.75), and sorted by ascending distance. Matched (x, y) coordinates are returned as `float32` arrays.
3. **Fundamental Matrix**: Estimated with `cv2.findFundamentalMat` (`FM_RANSAC`, threshold 3.0, confidence 0.99). The returned mask is used to keep only inlier point pairs.
4. **Rectification**: `cv2.stereoRectifyUncalibrated` computes homographies H_L, H_R that align epipolar lines horizontally. Both images are warped with `cv2.warpPerspective`. Black border regions from uncalibrated rectification are expected and handled by SGBM.
5. **Disparity (SGBM)**: A bilateral filter is applied to the rectified pair before matching to reduce noise in low-texture regions. The disparity is then computed with `cv2.StereoSGBM_create` (parameters in Table 1). Invalid disparities (< 0) are zeroed, and the result is normalised to $[0, 255]$ `uint8`.
6. **Visualisation**: A 1×3 Matplotlib figure shows the rectified pair and the colormapped disparity map (brighter $\approx$ closer).

Parameter	**Value**	**Reasoning**
`minDisparity`	0	Standard left-as-reference.
`numDisparities`	160	Wide depth range; divisible by 16.
`blockSize`	11	Balances detail and smoothness for this scene.
`P1`	968	$8 \cdot 1 \cdot 11^2$; small disparity change penalty.
`P2`	3872	$32 \cdot 1 \cdot 11^2$; larger discontinuity penalty.
`disp12MaxDiff`	1	Left/right consistency check.
`uniquenessRatio`	5	Relaxed; improves coverage on low-texture areas.
`speckleWindowSize`	200	Aggressive speckle noise filtering.
`speckleRange`	2	Tight variation limit inside a speckle.
`preFilterCap`	63	Normalises pixel intensities before matching.
`mode`	`SGBM_HH`	Full 8-direction dynamic programming.

Table 1: SGBM parameters. P_1, P_2 follow the recommended $8cb^2/32cb^2$ scaling (c=1 channel, b=11 block size).

Results

Figure 1 shows the rectified pair and disparity map. The central image region is well aligned and produces a coherent depth map; black border artefacts are inherent to uncalibrated rectification.

`output/disparity_results.png`

Figure 1: Rectified left, rectified right, and SGBM disparity map (brighter $\approx$ closer).